

Quantum Attacks & ECC

CSE 717 Information Security — Past-Paper Solutions (2024)

Source: Lc#6 (ECC), Lc#7 (Quantum Attacks) Yield: 1/5 years (2024 only), 5 marks

Algorithm Cheat Sheet

ECC Curve Definition

An elliptic curve over $\mathbb{F}_p$ is defined by:

$$y^2 \equiv x^3 + ax + b \pmod{p}$$

Each valid point (x, y) satisfies this equation mod p . The point at infinity $\mathcal{O}$ acts as the identity (0 in addition).

Point Doubling Formula — $2P$ where $P = Q = (x_1, y_1)$

Step 1: Compute slope λ

$$\lambda = \frac{3x_1^2 + a}{2y_1} \pmod{p}$$

Note: Division means multiply by the modular inverse. Use $ax \equiv 1 \pmod{p}$ to find a^{-1} .

Step 2: Compute x_3

$$x_3 = \lambda^2 - 2x_1 \pmod{p}$$

Step 3: Compute y_3

$$y_3 = \lambda(x_1 - x_3) - y_1 \pmod{p}$$

Result: $2P = (x_3, y_3)$

Negative mod trick: If result is negative, add multiples of p until positive.
 $(-a) \bmod p = (p - (a \bmod p)) \bmod p$

Point Addition Formula — $P + Q$ where $P \neq Q$, $P = (x_1, y_1)$, $Q = (x_2, y_2)$

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1} \pmod{p}$$

Then $x_3 = \lambda^2 - x_1 - x_2 \pmod{p}$, $y_3 = \lambda(x_1 - x_3) - y_1 \pmod{p}$

Quantum Attack Summary

Algorithm	Target	Impact
Shor's Algorithm	RSA (integer factorization)	Completely breaks RSA
Shor's Algorithm	ECC (discrete log, ECDLP)	Completely breaks ECC
Grover's Algorithm	Symmetric (AES, DES)	Halves key strength (quadratic speedup only)

Key distinction: Shor's gives *polynomial* speedup (exponential improvement) for public-key. Grover's gives only $\sqrt{\cdot}$ speedup (quadratic) for symmetric — manageable by doubling key size. AES-256 $\rightarrow$ still quantum-resistant. AES-128 $\rightarrow$ drops to ~ 64 -bit equivalent security.

2024 Exam (1/5 years — 5 marks total)

Q6(a) — 3 marks: ECC Point Doubling

2024 Q6(a) [3 marks]

Elliptic curve $E: y^2 = x^3 + 2x + 3$ over $\mathbb{F}_{19}$. Compute $2P$ for $P = (3, 6)$.

Solution — Point Doubling (Case B: $P = Q$)

Given: $x_1 = 3, y_1 = 6, a = 2, p = 19$

Step 1: Compute slope λ

$$\lambda = \frac{3x_1^2 + a}{2y_1} \pmod{19} = \frac{3(3)^2 + 2}{2(6)} \pmod{19} = \frac{27 + 2}{12} \pmod{19} = \frac{29}{12} \pmod{19}$$

Convert numerator: $29 \bmod 19 = 10$

Find $12^{-1} \bmod 19$: need $12k \equiv 1 \pmod{19}$

$$12 \times 8 = 96 = 5 \times 19 + 1 \implies 96 \equiv 1 \pmod{19}$$

So $12^{-1} \equiv 8 \pmod{19}$.

$$\lambda = 10 \times 8 = 80 \pmod{19} = 80 - 4(19) = 80 - 76 = \boxed{4}$$

Step 2: Compute x_3

$$x_3 = \lambda^2 - 2x_1 \pmod{19} = 4^2 - 2(3) \pmod{19} = 16 - 6 \pmod{19} = \boxed{10}$$

Step 3: Compute y_3

$$y_3 = \lambda(x_1 - x_3) - y_1 \pmod{19} = 4(3 - 10) - 6 \pmod{19} = 4(-7) - 6 \pmod{19} = -28 - 6 \pmod{19}$$

$$\text{Convert: } -34 + 2(19) = -34 + 38 = \boxed{4}$$

$$\therefore \mathbf{2P = (10, 4)}$$

Verification: $4^2 = 16$ and $10^3 + 2(10) + 3 = 1023 \equiv 16 \pmod{19} \checkmark$

Q6(b) — 2 marks: Why ECC/RSA More Vulnerable to Quantum

2024 Q6(b) [2 marks]

Why are ECC and RSA more vulnerable than symmetric algorithms to quantum attacks?

Solution

RSA and ECC are completely broken by Shor's Algorithm:

- RSA security relies on the **integer factorization problem** (factoring $n = pq$ is hard classically — exponential time).
- ECC security relies on the **Elliptic Curve Discrete Logarithm Problem (ECDLP)** (computing k from kP is hard classically).
- **Shor's Algorithm** (quantum) solves *both* integer factorization and discrete logarithm problems in **polynomial time** — completely breaking RSA and ECC.

Symmetric algorithms (AES) are only weakened, not broken:

- Symmetric crypto has no mathematical structure (factorization/discrete log) for Shor's to exploit.
- **Grover's Algorithm** can search an N -element space in $O(\sqrt{N})$ time — providing only a **quadratic speedup**.
- Effect: AES-128 $\rightarrow$ $\sim$ 64-bit equivalent security. This is manageable by **doubling key size**.
- AES-256 remains **quantum-resistant**.

Lecture Example (Lc#6 Slide 29–34) for Practice

Worked Example from Lecture: $y^2 \equiv x^3 + 2x + 2 \pmod{17}$

(3,6)+(3,6) — Point Doubling on a different curve/prime:

$x_1 = 3, y_1 = 6, a = 2, p = 17$

$$\lambda = \frac{3(9) + 2}{2(6)} \pmod{17} = \frac{29}{12} \pmod{17}$$

$29 \bmod 17 = 12$ and $12^{-1} \bmod 17 = 10$ (since $12 \times 10 = 120 \equiv 1 \pmod{17}$)

$$\lambda = 12 \times 10 = 120 \bmod 17 = 1$$

$$x_3 = 1 - 6 = -5 \bmod 17 = 12 \quad y_3 = 1(3 - 12) - 6 = -15 \bmod 17 = 2$$

Result: $(3, 6) + (3, 6) = (12, 2)$ (matches lecture slide 34)

Note: The exam uses $b = 3, p = 19$ while the lecture uses $b = 2, p = 17$ — same method, different numbers.

Exam Strategy

What to Write in 5 Minutes

Part	What to write
Q6(a) [3 marks]	Write: Given $P = (3, 6)$, $a = 2$, $p = 19$. Show Steps 1–3 with every intermediate mod. Box the answer $2P = (10, 4)$.
Q6(b) [2 marks]	One sentence each for RSA (Shor breaks factorization), ECC (Shor breaks ECDLP), symmetric (Grover only halves, AES-256 safe).

Watch out:

- Use $x_3 = \lambda^2 - 2x_1$ for doubling (NOT $\lambda^2 - x_1 - x_2$, which is for $P \neq Q$ addition).
- Always convert negative results mod p by adding p repeatedly.
- For $a^{-1} \bmod p$: find k such that $ak \equiv 1 \pmod{p}$ by trial or Extended Euclidean.